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AI CAREERS FOR WOMEN

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Pay-Lens: visualizing salary truth with AI & BI

Project Domain

[Excel/power BI with copilot & chatbot]

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

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Year - 2026

3. Problem Statement

Background of the Chosen Domain and Data-Related Challenges

The project focuses on Human Resources (HR) and Payroll Management, a domain where organizations handle vast amounts of sensitive compensation data. Currently, this data is often raw, inconsistent, or locked in complex payroll systems. A primary challenge is the Information Asymmetry between the employer and the employee; while companies use advanced tracking, employees—especially gig and blue-collar workers—lack clear, verifiable access to how their wages are calculated.

Significance in a Real-World Context

This problem is critical because it involves Legal Compliance and Financial Stability. Under Indian labour laws (such as the Factories Act), overtime must be paid at 2x the ordinary rate. In the real world, poor transparency often leads to systemic underpayment of overtime or incorrect statutory deductions (PF, Professional Tax) that go unnoticed by the workforce. Furthermore, pay inequity—where similar roles have widely different packages—erodes organizational trust and fairness.

Specific Gap or Inefficiency Addressed

The project aims to bridge the **"Transparency and Skills Gap"**:

- **The Transparency Gap:** There is a lack of clear tools to help employees understand the conversion from Gross CTC to In-hand salary.
- **The Skills Gap:** Most HR professionals lack the Prompt Engineering skills needed to use AI tools like Microsoft Copilot to automate the detection of payroll discrepancies.
- **The Analytical Gap:** There is often no automated "Ratio Analysis & interpretation" to compare internal salaries against Market Median benchmarks.

Who is Affected and the Value Delivered

- **Affected Parties:** Blue-collar workers and gig employees face the highest risk of wage loss. HR departments are affected by high turnover costs, which typically equal 1.5x the annual salary of the departing employee.
- **Value Delivered:** The solution provides Automated Verification to ensure legal compliance, reducing the risk of lawsuits and penalties. It delivers Financial Literacy to employees through clear deduction breakdowns, improving morale and retention

4. Objectives

- To collect, explore, clean and preprocess the raw payroll dataset (1,000 records) by handling values standardizing it to understand salary structures and data quality.

- To perform Ratio Analysis by comparing internal salaries with Market Median benchmarks (50th percentiles).

- To build an Interactive Power BI Dashboard that visualizes the relationship between salary increases, tenure, and legal compliance.

- To build an agent which is able to generate business report in less than 5 minutes, analyse any discrepancy, suggestion to improve the flaws in payroll system. (integration through subscription)

- To build a chatbot to bridge transparency and skill gap for Employees and Employer.
- To build an Web application to add an extension to my project.

5. Methodology / Implementation Plan

sr.no	Phase	Implementation Details
1	Data Collection & Exploration	Obtain the 1,000-record payroll dataset from Kaggle including Employee Master, Attendance, and Market Benchmark tables. We will identify key fields such as CTC_Annual, Market_Median_Salary, and Overtime Hours to understand the baseline for transparency.
2	Data Cleaning & Preprocessing	Use Excel to handle values in Employee_ID, fix date formats for Date_of_Joining, and standardize text. For the AI component, we will draft structured.
3	Analysis	Implement Excel formulas and DAX measures in Power BI to calculate the Market Positioning Index and verify 2x Overtime pay compliance. For financial literacy part free AI tools can be integrated
4	Visualization / Output	Develop an interactive Power BI dashboard featuring KPI cards for "Total Payroll Cost" and "Salary vs. Market Ratio." The output will include slicers to filter by department (e.g., IT, Sales) to identify specific areas of pay inequity.
5	Web application	Embed the Pay Lens Paroll dashboard inside streamlit and convert it into a decision support interface for compensation correction

6	Documentation & Submission	Compile the final project report, document all AI prompts used, and upload the complete repository to GitHub, azure portal. A video demonstration will be recorded to showcase the dashboard's ability to flag legal non-compliance.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i5-12400/AMD Ryzen 5 5500U or higher, minimum DDR5 8 GB RAM (DDR5 16 GB recommended)
- Operating System: Windows 11 or macOS 13 Ventura
- Microsoft Excel (Microsoft 365 / Excel 2019 or higher)
- Power BI Desktop — latest version (free download from Microsoft)
- Web browser — Google Chrome or Microsoft Edge (latest version)
- Vs studio – python enabled in it 3.12 version

B. Dataset

1) 1,000-record CSV containing employee IDs, Department, CTC, and Market Benchmarks. (multiple datasets sourced from Kaggle and rest of the column created by excel financial formulas)
2) https://www.kaggle.com/datasets?search=employee
3) After collecting different data set from Kaggle following steps need to done:
   a) Define the columns required
   b) Apple Excel formulas like
      i) To create a unique employee ID: ="EMP"&TEXT(ROW(A1),"000")
      ii) To allot the dept: create a department list in worksheet -- use a formula given below by selecting the dept list reference
      =INDEX(Z1: Z6,RANDBETWEEN(1,6))
      iii) To allot random date of joining:
      =RANDBETWEEN(DATE(2019,1,1),DATE(2025,12,31))
      iv) To calculate CTC and all the financial calculation can use following formulas
      =E2*12% [employee PF]
      =E2*40% [HRA]
      =E2*30% [allowances]
      =SUM(E2:H2) [total CTC]
      =I2*12 [annual CTC]
      Note: these all formulas are on the basis of my dataset and I have created the dataset using this formula.

C. Project Timeline

A table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*
2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Expected Outcomes / Conclusion

- **Intelligent Payroll Data Infrastructure:**
  Development of a clean, standardized, and audit-ready payroll dataset (1,000+ records), enabling reliable financial analysis and decision-making within HR systems.

- **Automated Legal Alerts:** A system that successfully flags any underpayment of **Overtime (2x rate)**, ensuring compliance with Indian Labor Laws (Factories Act).

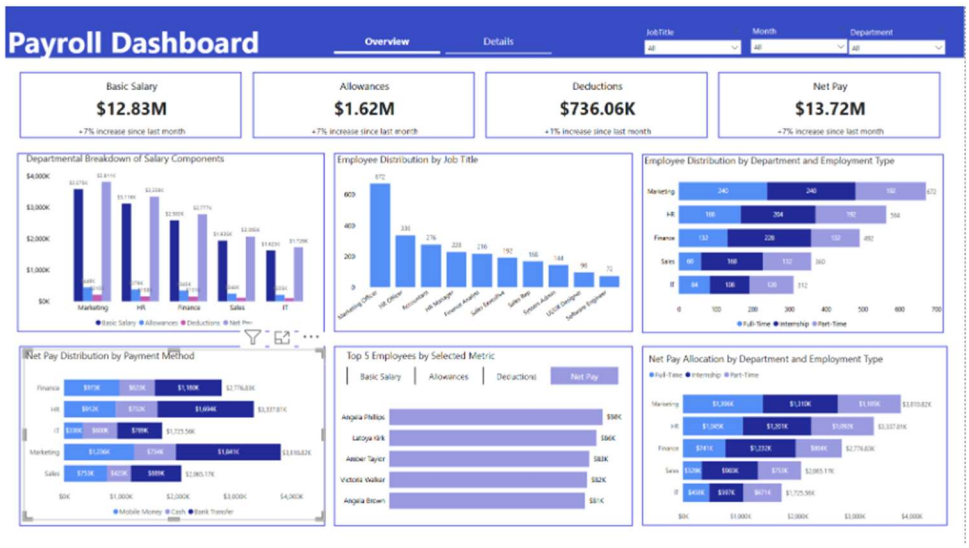
- **Quantitative Pay Equity Framework:**
  Creation of a structured **Market Positioning Index** to evaluate internal salary distributions against industry benchmarks, enabling organizations to detect pay disparities and optimize compensation strategies.

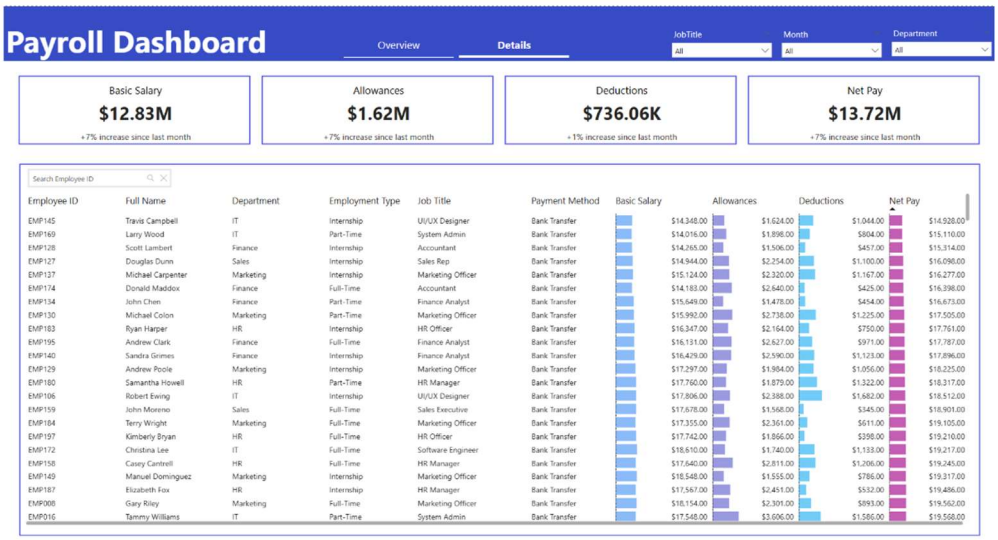
- **Enhanced Financial Transparency for Workforce:**
  Transformation of complex salary structures into **simplified, AI-assisted explanations**, empowering employees—especially gig and blue-collar workers—with clarity on deductions, benefits, and net pay.

- **Decision-Intelligence Dashboard:**
  Development of an interactive Power BI dashboard that converts raw payroll data into **actionable insights**, highlighting trends in compensation, tenure, departmental cost structures, and potential attrition risks.

- **AI-Augmented HR Analytics Capability:**
  Integration of prompt-based AI tools to assist HR professionals in detecting anomalies, interpreting financial ratios, and automating repetitive analytical tasks, thereby reducing manual effort and improving accuracy.

- **Bridging the Transparency–Analytics Gap:**
  The project establishes a scalable framework that combines **data analytics, AI, and business intelligence** to address systemic inefficiencies in payroll systems and improve organizational trust.





Note: these are the visuals taken from the GitHub, link is mentioned in the references. After doing all the task on dataset, my dashboard will look like this above-mentioned dashboard.

8. References

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5. The Gig Workers' Strike Shows that Sustained Pressure from the Labor Force Does Yield Results - The Wire [an article]

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